

@jridgewell/trace-mapping Trace the original position through a source map trace-mapping allows you to take the line and column of an output file and trace it to the original location in the source file through a source map. You may already be familiar with the source-map package's SourceMapConsumer. 1	This provides the same originalPositionFor and generatedPositionFor API, without requiring WASM. Installation npm install @jridgewell/trace-mapping Usage import { TraceMap, 2	originalPositionFor, generatedPositionFor, sourceContentFor, isIgnored, } from '@jridgewell/trace-mapping'; const tracer = new TraceMap({ version: 3, sources: ['input.js'], sourcesContent: ['content of input.js'], names: ['foo'], 3	mappings: 'KAYCIA', ignoreList: [], }); // Lines start at line 1, columns at column 0. const traced = originalPositionFor(tracer, { line: 1, column: 5 })); assert.deepEqual(traced, { source: 'input.js', line: 42, 4
SectionedSourceMaps The sourcemap spec defines a special sections field that's designed to handle // Segments are [outputColumn, sourceIndex, sourceLine, sourceColumn, namesIndex] // Again, line is 0-base and so is sourceLine assert.deepEqual(traced, [5, 0, 41, 4, 0]); 8	We also provide a lower level API to get the actual segment that matches our line and column. Unlike originalPositionFor, traceSegment uses a 0-base for line: import { traceSegment } from '@jridgewell/trace-mapping'; // line is 0-base. const traced = traceSegment(tracer, /* line */ 0, /* column */ 5); 7	source: 'input.js', line: 42, column: 4, assert.deepEqual(generated, { line: 1, column: 5, }); const ignored = isIgnored(tracer, 'input.js'); assert.equal(ignored, false); 6	column: 4, name: 'foo', }); const content = sourceContentFor(tracer, traced.source); assert.strictEqual(content, 'content for input.js'); const generated = generatedPositionFor(tracer, { 5
concatenation of output code with associated sourcemaps. This type of sourcemap is rarely used (no major build tool produces it), but if you are hand coding a concatenation you may need it. We provide an AnyMap helper that can receive either a regular sourcemap or a SectionedSourceMap and returns a TraceMap instance: import { AnyMap } from '@jridgewell/trace-mapping'; const fooOutput = 'foo'; const barOutput = 'bar'; 6	const output = [fooOutput, barOutput].join('\n'); const sectioned = new AnyMap({ version: 3, sections: [{ // 0-base line and column offset: { line: 0, column: 0 }, // fooOutput's sourcemap map: { version: 3, 10	sources: ['foo.js'], names: ['foo'], mappings: 'AAAAA', }, }, { // barOutput's sourcemap will not affect the first line, only the second offset: { line: 1, column: 0 }, map: { version: 3, 11	sources: ['bar.js'], names: ['bar'], mappings: 'AAAAA', }, }, }, const traced = originalPositionFor(sectioned, { line: 2, 12
trace-mapping: encoded JS/N input x 405 ops/sec ±1.47% (88 runs sampled) trace-mapping: decoded JS/N input x 458 ops/sec ±1.63% (91 runs sampled) source-map-js: encoded JS/N input x 75.48 ops/sec ±1.64% (67 runs sampled) source-map-0.6.1: encoded JS/N input x 39.37 ops/sec ±1.44% (53 runs sampled) 16	source-map-0.6.1 bytes source-map-0.8.0 bytes Chrome dev tools 8672912 Smallest memory usage is trace-mapping bytes trace-mapping: encoded JS/N input x 405 ops/sec ±1.47% (88 runs sampled) trace-mapping: decoded JS/N input x 458 ops/sec ±1.63% (91 runs sampled) source-map-js: encoded JS/N input x 75.48 ops/sec ±1.64% (67 runs sampled) source-map-0.6.1: encoded JS/N input x 39.37 ops/sec ±1.44% (53 runs sampled) 15	Benchmarks node v20.10.0 amp.js.map - 45120 segments Memory Usage: 414164 bytes trace-mapping decoded 6274352 bytes source-map-js 10968904 bytes 14	column: 0, }); assert.deepEqual(traced, { source: 'bar.js', line: 1, column: 0, name: 'bar', }); 13

sampled) Chrome dev tools: encoded Object input x 150 ops/sec ±1.76% (79 runs sampled) Fastest is trace-mapping: decoded Object input Trace speed (random): trace-mapping: decoded originalPositionFor x 44,946 ops/sec ±0.16% (99 runs sampled) trace-mapping: encoded originalPositionFor x 37,995 ops/sec	±1.81% (89 runs sampled) source-map-js: encoded originalPositionFor x 9,230 ops/sec ±1.36% (93 runs sampled) source-map-0.6.1: encoded originalPositionFor x 8,057 ops/sec ±0.84% (96 runs sampled) source-map-0.8.0: encoded originalPositionFor x 28,198 ops/sec ±1.12% (91 runs sampled) Chrome dev tools: encoded originalPositionFor x 46,276 ops/sec	±1.35% (95 runs sampled) Fastest is Chrome dev tools: encoded originalPositionFor Trace speed (ascending): trace-mapping: decoded originalPositionFor x 204,406 ops/sec ±0.19% (97 runs sampled) trace-mapping: encoded originalPositionFor x 196,695 ops/sec ±0.24% (99 runs sampled) source-map-js: encoded	originalPositionFor x 11,948 ops/sec ±0.94% (99 runs sampled) source-map-0.6.1: encoded originalPositionFor x 10,730 ops/sec ±0.36% (100 runs sampled) source-map-0.8.0: encoded originalPositionFor x 51,427 ops/sec ±0.21% (98 runs sampled) Chrome dev tools: encoded originalPositionFor x 162,615 ops/sec ±0.18% (98 runs sampled) Fastest is trace-mapping: decoded
17	18	19	20
24 Trace speed (random): trace-mapping: decoded Fastest is trace-mapping: decoded Object input Chrome dev tools: encoded Object input x 22.14 ops/sec ±3.79% (41 runs sampled) source-map-0.6.1: encoded Object input x 4.23 ops/sec ±3.43% (15 runs sampled)	23 Init speed: trace-mapping: decoded J50N input x 17.82 ops/sec ±6.35% (35 runs sampled) trace-mapping: encoded J50N input x 31.57 ops/sec ±7.50% (43 runs sampled) trace-mapping: decoded Object input x 867 ops/sec ±0.74% (94 runs sampled) trace-mapping: encoded Object input x 33.83 ops/sec ±7.66% (46 runs sampled) source-map-js: encoded Object input x 6.58 ops/sec ±3.31% (20 runs sampled)	22 bytes source-map-js 51676808 bytes source-map-0.6.1 63367136 bytes source-map-0.8.0 43158400 bytes Chrome dev tools 50721552 bytes Smallest memory usage is trace-mapping decoded	21 originalPositionFor *** babel.min.js.map - 347793 segments Memory Usage: trace-mapping decoded 18504 bytes trace-mapping encoded 35428008
originalPositionFor x 78,234 ops/sec ±1.48% (29 runs sampled) trace-mapping: encoded originalPositionFor x 60,761 ops/sec ±1.35% (21 runs sampled) source-map-js: encoded originalPositionFor x 51,448 ops/sec ±2.17% (89 runs sampled) source-map-0.6.1: encoded originalPositionFor x 47,221 ops/sec ±1.99% (15 runs sampled) source-map-0.8.0: encoded	originalPositionFor x 84,002 ops/sec ±1.45% (27 runs sampled) Chrome dev tools: encoded originalPositionFor x 106,457 ops/sec ±1.38% (37 runs sampled) Fastest is Chrome dev tools: encoded originalPositionFor Trace speed (ascending): trace-mapping: decoded originalPositionFor x 930,943 ops/sec ±0.25% (99 runs sampled)	trace-mapping: encoded originalPositionFor x 843,545 ops/sec ±0.34% (97 runs sampled) source-map-js: encoded originalPositionFor x 114,510 ops/sec ±1.37% (36 runs sampled) source-map-0.6.1: encoded originalPositionFor x 87,412 ops/sec ±0.72% (92 runs sampled) source-map-0.8.0: encoded originalPositionFor x 197,709 ops/sec ±0.86% (59 runs sampled)	Chrome dev tools: encoded originalPositionFor x 688,983 ops/sec ±0.33% (98 runs sampled) Fastest is trace-mapping: decoded originalPositionFor *** preact.js.map - 1992 segments
25	26	27	28
32 Trace speed (random): trace-mapping: decoded originalPositionFor x 91.028 ops/sec ±0.14% (94 runs sampled) trace-mapping: encoded originalPositionFor x 84,348 ops/sec ±0.26% (98 runs sampled) source-map-js: encoded Object input Fastest is trace-mapping: decoded sampled)	31 sampled) trace-mapping: encoded Object input x 14,980 ops/sec ±0.28% (100 runs sampled) source-map-js: encoded Object input x 2,544 ops/sec ±0.16% (99 runs sampled) source-map-0.6.1: encoded Object input x 1,221 ops/sec ±0.37% (97 runs sampled) Chrome dev tools: encoded Object input x 4,241 ops/sec ±0.39% (93 runs sampled)	30 bytes Smallest memory usage is trace-mapping decoded Init speed: trace-mapping: decoded J50N input x 3,709 ops/sec ±0.13% (99 runs sampled) trace-mapping: encoded J50N input x 6,447 ops/sec ±0.22% (101 runs sampled) trace-mapping: decoded Object input x 83,062 ops/sec ±0.23% (100 runs sampled)	29 Memory Usage: trace-mapping decoded 33136 bytes trace-mapping encoded 254240 bytes source-map-js 837488 bytes source-map-0.6.1 961928 bytes source-map-0.8.0 54384 bytes Chrome dev tools 709680

originalPositionFor x 26,998 ops/sec ±0.23% (98 runs sampled) source-map-0.6.1: encoded originalPositionFor x 18,049 ops/sec ±0.26% (100 runs sampled) source-map-0.8.0: encoded originalPositionFor x 41,916 ops/sec ±0.28% (98 runs sampled) Chrome dev tools: encoded originalPositionFor x 88,616 ops/sec ±0.14% (98 runs sampled) Fastest is trace-mapping: decoded	originalPositionFor Trace speed (ascending): trace-mapping: decoded originalPositionFor x 319,960 ops/sec ±0.16% (100 runs sampled) trace-mapping: encoded originalPositionFor x 302,153 ops/sec ±0.18% (100 runs sampled) source-map-js: encoded originalPositionFor x 35,574 ops/sec ±0.19% (100 runs sampled)	source-map-0.6.1: encoded originalPositionFor x 19,943 ops/sec ±0.12% (101 runs sampled) source-map-0.8.0: encoded originalPositionFor x 54,648 ops/sec ±0.20% (99 runs sampled) Chrome dev tools: encoded originalPositionFor x 278,319 ops/sec ±0.17% (102 runs sampled) Fastest is trace-mapping: decoded originalPositionFor	*** react.js.map - 5726 segments Memory Usage: <table><tr><td>trace-mapping decoded</td><td>10872</td></tr><tr><td>bytes</td><td></td></tr><tr><td>trace-mapping encoded</td><td>681512</td></tr><tr><td>bytes</td><td></td></tr><tr><td>source-map-js</td><td>2563944</td></tr></table>	trace-mapping decoded	10872	bytes		trace-mapping encoded	681512	bytes		source-map-js	2563944
trace-mapping decoded	10872												
bytes													
trace-mapping encoded	681512												
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source-map-js	2563944												
33	34	35	36										
40	39	38	37										
originalPositionFor x 579,548 ops/sec ±0.35% (97 runs sampled) source-map-js: encoded originalPositionFor x 230,983 ops/sec ±0.62% (54 runs sampled) source-map-0.6.1: encoded originalPositionFor x 158,145 ops/sec ±0.80% (46 runs sampled) source-map-0.8.0: encoded originalPositionFor x 343,801 ops/sec ±0.55% (96 runs sampled) Chrome dev tools: encoded	originalPositionFor x 620,201 ops/sec ±0.33% (96 runs sampled) trace-mapping: encoded Trace speed (random): trace-mapping: decoded originalPositionFor x 620,201 ops/sec ±0.33% (96 runs sampled) trace-mapping: decoded Fastest is trace-mapping: decoded sampled originalPositionFor x 1,524 ops/sec ±0.57% (92 runs sampled) Chrome dev tools: encoded originalPositionFor x 418 ops/sec ±0.33% (94 runs sampled)	originalPositionFor x 1,887 ops/sec ±0.28% (99 runs sampled) trace-mapping: encoded originalPositionFor x 4,749 ops/sec ±0.48% (97 runs sampled) source-map-0.8.0: decoded originalPositionFor x 74,236 ops/sec ±0.11% (99 runs sampled) trace-mapping: encoded originalPositionFor x 5,752 ops/sec ±0.38% (100 runs sampled) source-map-js: encoded originalPositionFor x 806 ops/sec ±0.19% (97 runs sampled) source-map-0.6.1: encoded originalPositionFor x 1,887 ops/sec ±0.28% (99 runs sampled)	bytes source-map-0.6.1 2150864 bytes source-map-0.8.0 88680 bytes Chrome dev tools 1149576 Smallest memory usage is trace-mapping bytes decoded Init speed: trace-mapping: decoded originalPositionFor x 1,887 ops/sec ±0.28% (99 runs sampled)										
originalPositionFor x 659,649 ops/sec ±0.49% (98 runs sampled) Fastest is Chrome dev tools: encoded originalPositionFor Trace speed (ascending): trace-mapping: decoded originalPositionFor x 2,368,079 ops/sec ±0.32% (98 runs sampled) trace-mapping: encoded originalPositionFor x 2,134,039 ops/sec ±2.72% (87 runs sampled)	source-map-js: encoded originalPositionFor x 290,120 ops/sec ±2.49% (82 runs sampled) source-map-0.6.1: encoded originalPositionFor x 187,613 ops/sec ±0.86% (49 runs sampled) source-map-0.8.0: encoded originalPositionFor x 479,569 ops/sec ±0.65% (96 runs sampled) Chrome dev tools: encoded originalPositionFor x 2,048,414 ops/sec ±0.24% (98 runs sampled)	Fastest is trace-mapping: decoded originalPositionFor *** vscode.map - 2141001 segments Memory Usage: <table><tr><td>trace-mapping decoded</td><td>5206584</td></tr><tr><td>bytes</td><td></td></tr></table>	trace-mapping decoded	5206584	bytes		trace-mapping encoded 208370336 bytes source-map-js 278493008 bytes source-map-0.6.1 391564048 bytes source-map-0.8.0 257508787 bytes Chrome dev tools 291053000 bytes Smallest memory usage is trace-mapping decoded						
trace-mapping decoded	5206584												
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41	42	43	44										
48	47	46	45										
originalPositionFor x 107,758 ops/sec ±1.94% (45 runs sampled) Chrome dev tools: encoded originalPositionFor x 188,550 ops/sec ±1.85% (79 runs sampled) Fastest is trace-mapping: decoded originalPositionFor Trace speed (ascending): trace-mapping: decoded originalPositionFor x 447,621 ops/sec ±3.64% (94 runs sampled)	originalPositionFor x 257,019 ops/sec ±0.97% (93 runs sampled) trace-mapping: encoded originalPositionFor x 179,163 ops/sec ±0.83% (92 runs sampled) source-map-js: encoded originalPositionFor x 73,337 ops/sec ±1.35% (87 runs sampled) source-map-0.6.1: encoded originalPositionFor x 38,797 ops/sec ±1.66% (88 runs sampled) source-map-0.8.0: encoded	originalPositionFor x 1,07 ops/sec ±13.84% (7 runs sampled) source-map-0.6.1: encoded originalPositionFor x 0.60 ops/sec ±2.43% (6 runs sampled) Chrome dev tools: encoded originalPositionFor x 2.61 ops/sec ±22.00% (11 runs sampled) Fastest is trace-mapping: decoded sampled originalPositionFor x 103 ops/sec ±0.93% (77 runs sampled) trace-mapping: decoded originalPositionFor x 3.29 ops/sec ±36.13% (13 runs sampled) trace-mapping: encoded originalPositionFor x 5.42 ops/sec ±28.54% (19 runs sampled) source-map-js: encoded originalPositionFor x 1,07 ops/sec ±13.84% (7 runs sampled)	Init speed: trace-mapping: decoded originalPositionFor x 1.63 ops/sec ±33.88% (9 runs sampled) trace-mapping: encoded originalPositionFor x 0.60 ops/sec ±2.43% (6 runs sampled) Chrome dev tools: encoded originalPositionFor x 2.61 ops/sec ±22.00% (11 runs sampled) Fastest is trace-mapping: decoded sampled Trace speed (random): trace-mapping: decoded										

trace-mapping: encoded originalPositionFor x 323,698 ops/sec ±5.20% (88 runs sampled) source-map-js: encoded originalPositionFor x 78,387 ops/sec ±1.69% (89 runs sampled) source-map-0.6.1: encoded originalPositionFor x 41,016 ops/sec ±3.01% (25 runs sampled) source-map-0.8.0: encoded originalPositionFor x 124,204 ops/sec ±0.90% (92 runs sampled) 49	Chrome dev tools: encoded originalPositionFor x 230,087 ops/sec ±2.61% (93 runs sampled) Fastest is trace-mapping: decoded originalPositionFor 50	51	52
95	55	54	53
57	58	59	60
49	69	79	19